LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	29
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and explanation of altitude hold and airspeed hold
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and explanation of altitude hold and airspeed hold.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS mode selection on simulator; Instructor supervises mode engagement and verifies parameter stability.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Which sensors provide data for altitude hold?
(c) Why is airspeed hold critical during approach?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of Altitude Hold Mode • AFCS Logic for Altitude Maintenance • Pitch Control Authority in Altitude Hold • Altitude Hold Engagement Criteria • Altitude Hold Disengagement Conditions	L & Ds	PPT, IB	15 min
(b)	• Purpose of Airspeed Hold Mode • Thrust Management in Airspeed Hold • Pitch Control Authority in Airspeed Hold • Airspeed Hold Engagement Criteria • Airspeed Hold Disengagement Conditions	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Altitude and Airspeed Hold • Mode Transition Logic • Instructor Demonstration on Simulator • Trainee Observation and Verification • Safety Precautions for Mode Changes	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS maintains altitude without pilot input.
(b) Describe the difference in control authority between altitude hold and airspeed hold.
(c) What happens to pitch and thrust when airspeed hold is engaged?

12. Home Assignment and Follow Up	Prepare a brief note on how the need for air data computers supports the accuracy of altitude and airspeed hold functions, referencing LP 30.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Vertical speed hold and heading hold operation, Mach hold and need for air data computers.	Refer to the master syllabus reference list	LP 30

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min